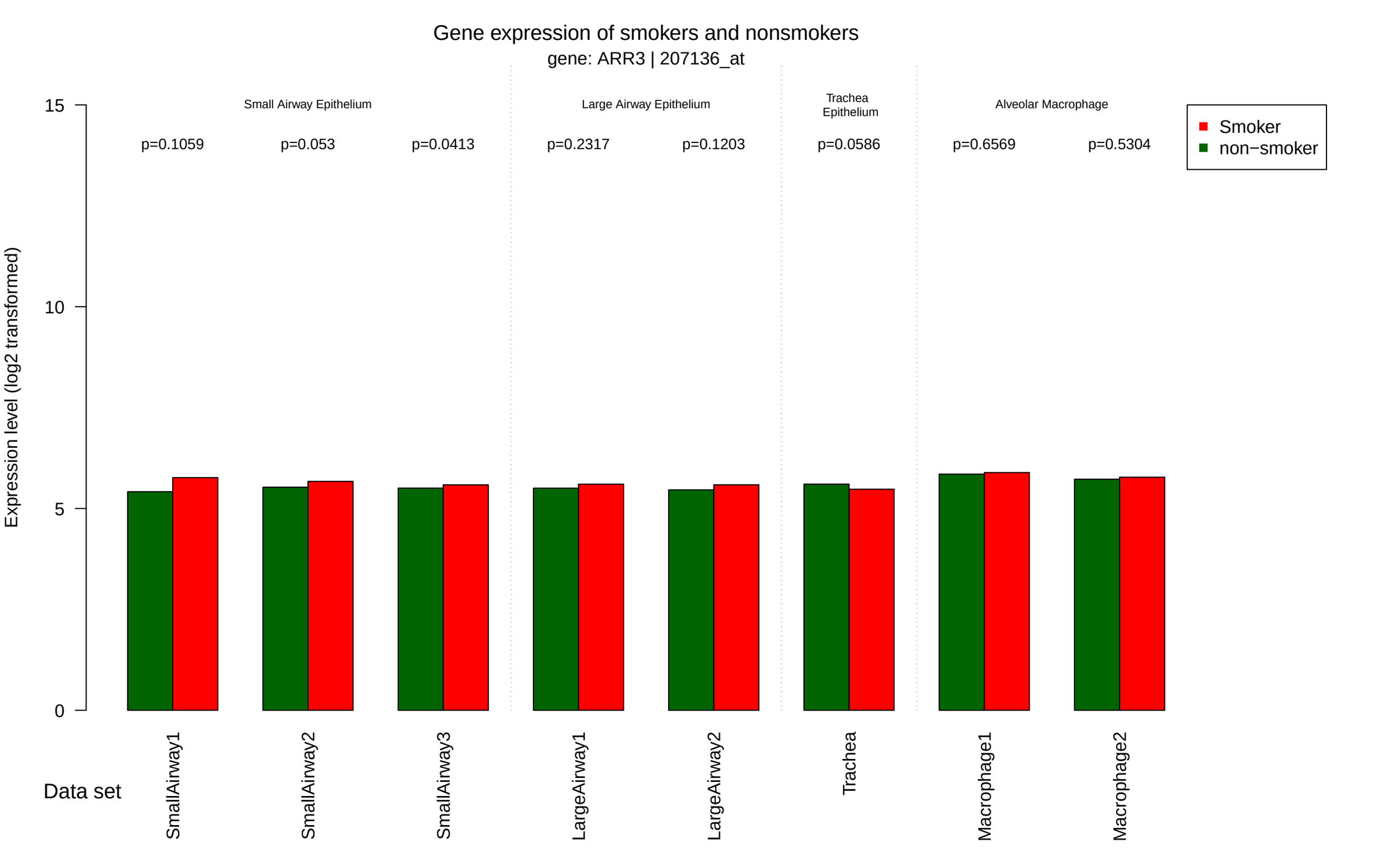
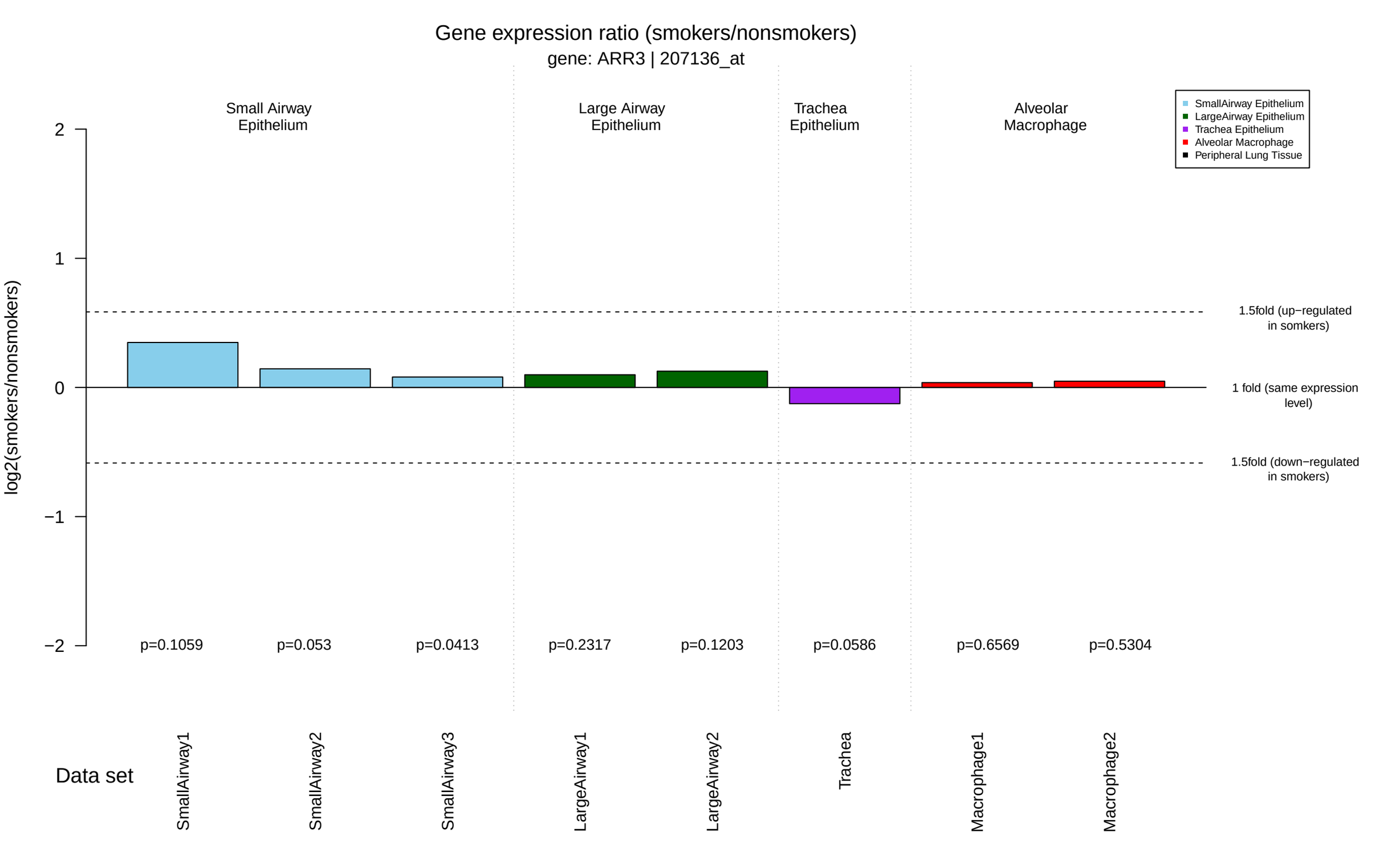
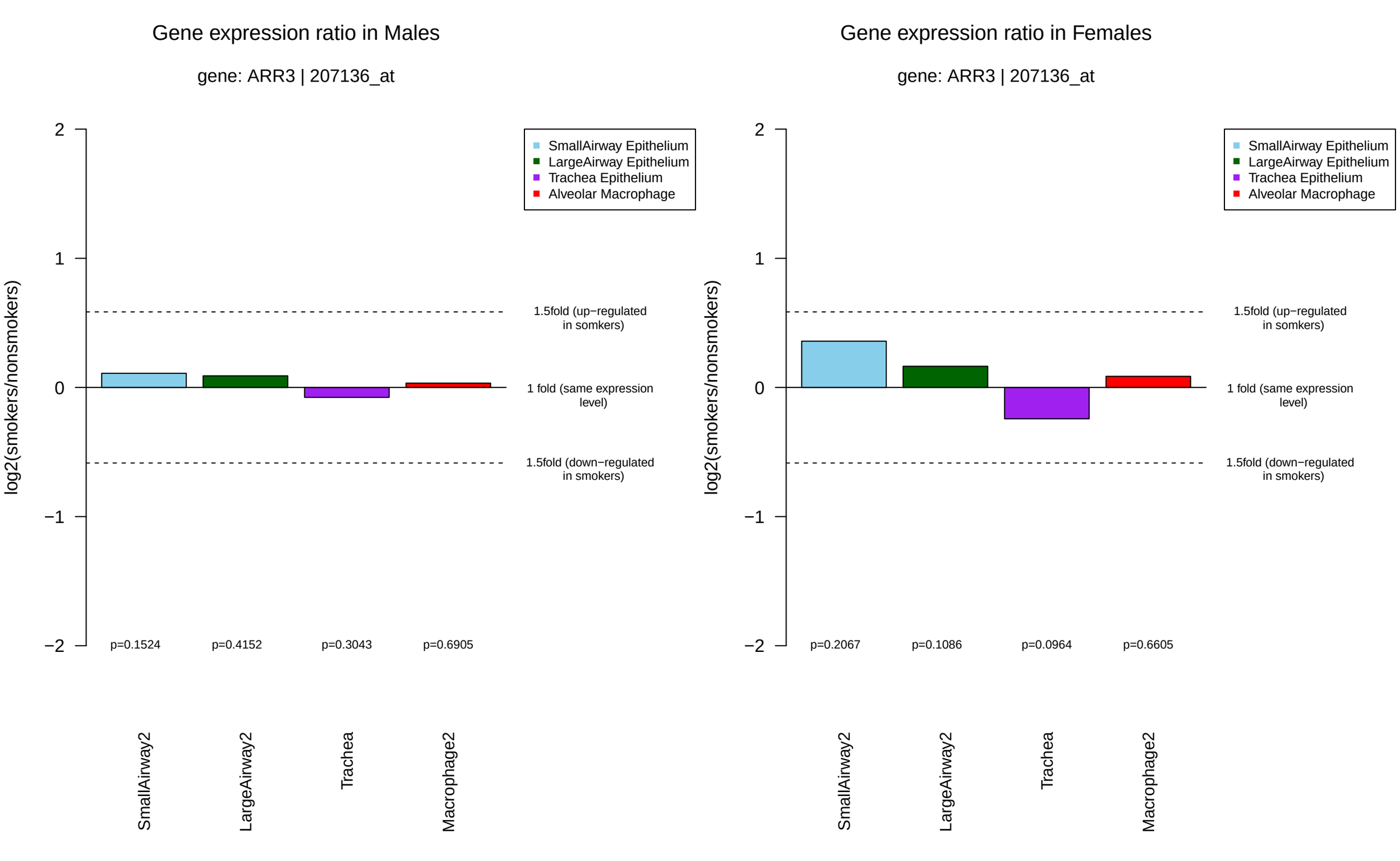
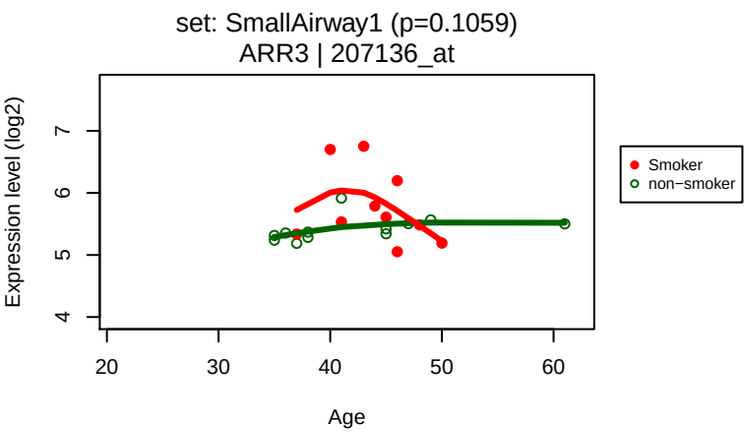
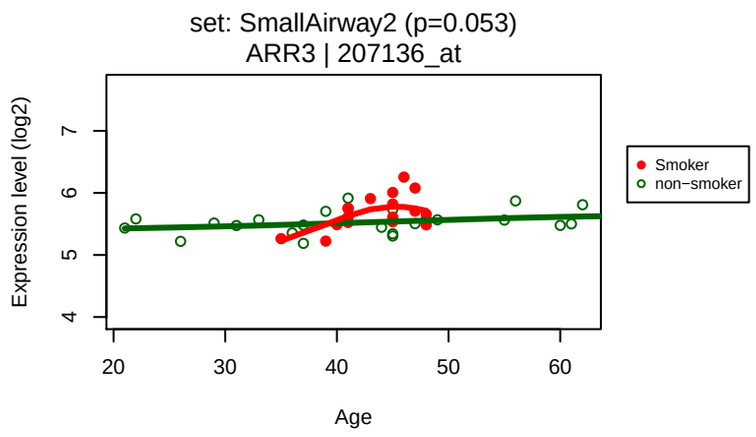
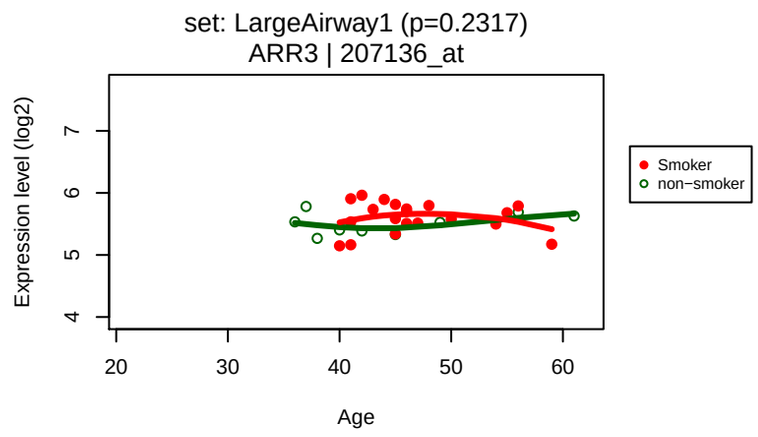

Gene expression of smokers and nonsmokers
gene: ARR3 | 207136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1059
p=0.053
p=0.0413
p=0.2317
p=0.1203
p=0.0586
p=0.6569
p=0.5304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

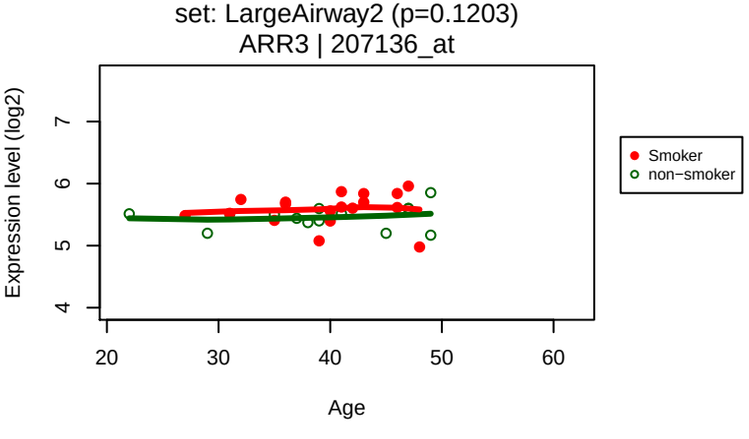

Gene expression ratio (smokers/nonsmokers)
gene: ARR3 | 207136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.053
p=0.0413
p=0.2317
p=0.1203
p=0.0586
p=0.6569
p=0.5304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

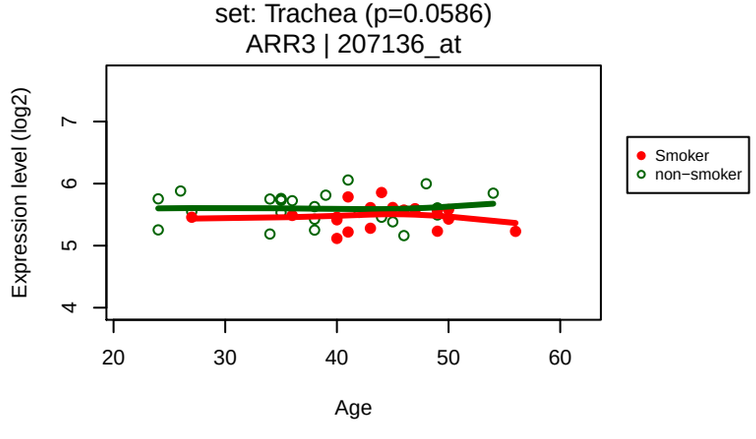

Gene expression ratio in Males
gene: ARR3 | 207136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1524
p=0.4152
p=0.3043
p=0.6905
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARR3 | 207136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2067
p=0.1086
p=0.0964
p=0.6605
SmallAirway2
LargeAirway2
Trachea
Macrophage2

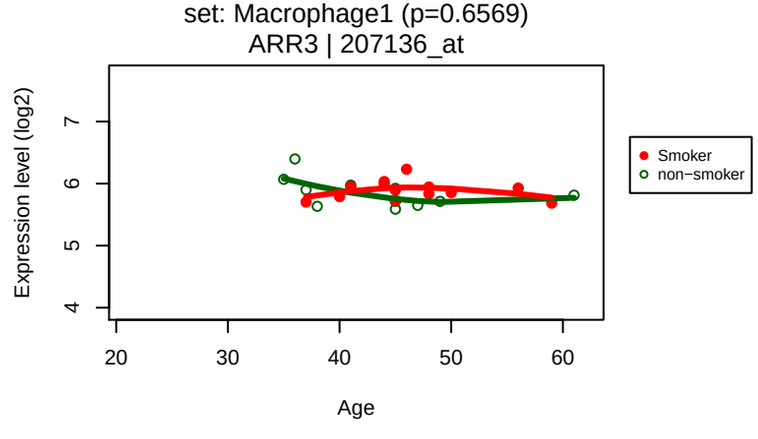

set: SmallAirway1 (p=0.1059)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker

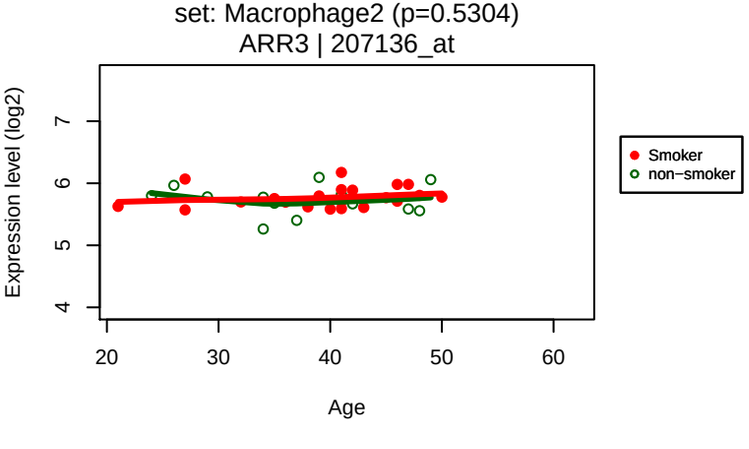

set: SmallAirway2 (p=0.053)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker

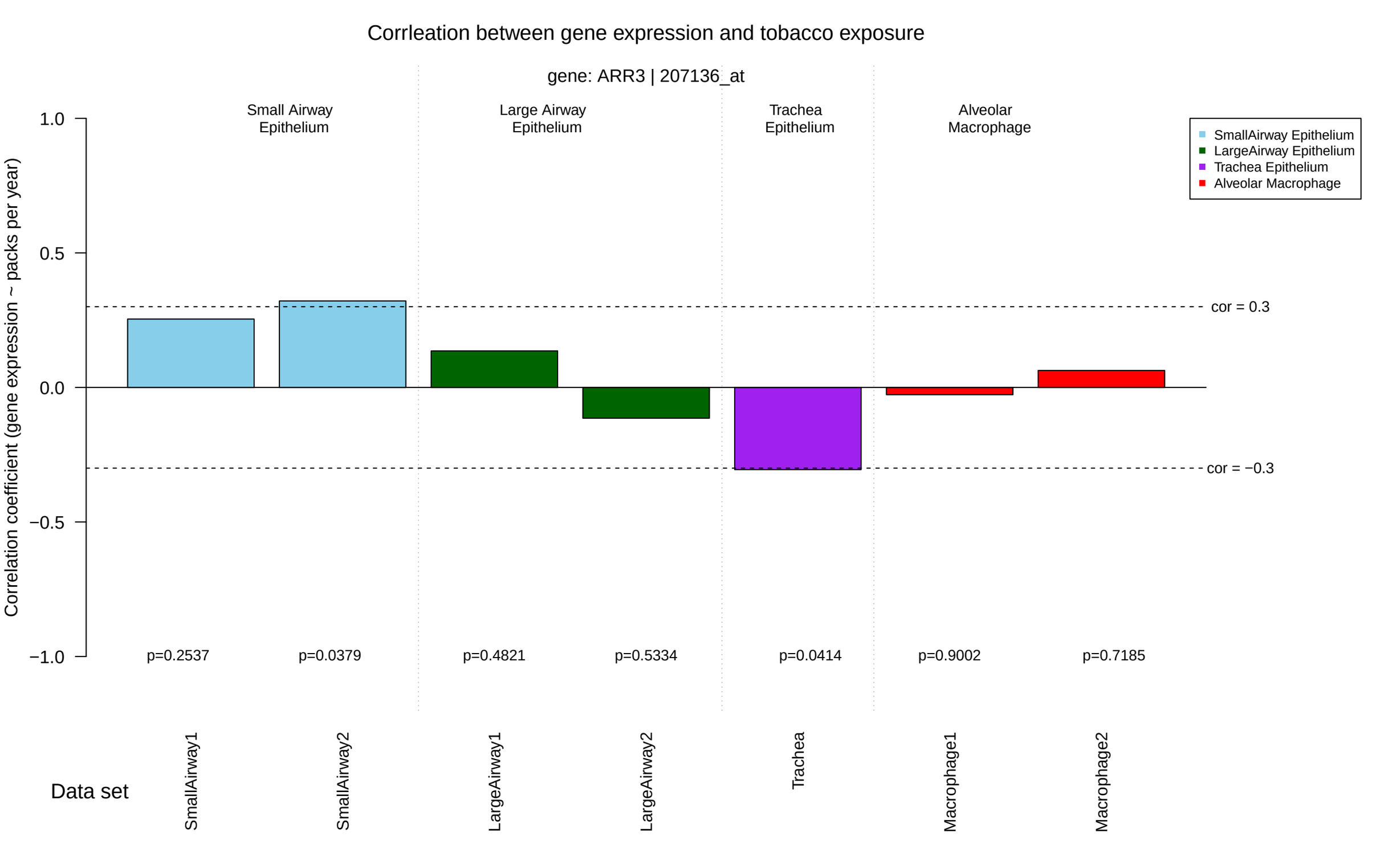

set: LargeAirway1 (p=0.2317)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1203)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0586)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6569)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5304)
ARR3 | 207136_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARR3 | 207136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2537
p=0.0379
p=0.4821
p=0.5334
p=0.0414
p=0.9002
p=0.7185
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2